

कवच

Kavach

The offline scam-call shield
that catches the **script**, not
the voice — and **speaks up**
before the money's gone.

On-device

No internet permission

12 languages

Runs on a ₹6,000 phone

Beyond Tomorrow Summit 2026 · Anbuchelvan Ganesan · github.com/Anbu-00001/Kavach

THE PROBLEM

The most expensive everyday crime — aimed at people who can't fight back

₹19,000cr +206%

lost to the "digital arrest" scam in India, 2025

cyber-fraud losses, 2024 vs 2023 (I4C)

60%

of 2024 victims were making their *first* digital payment

AI voice cloning removed the last tell. Scammers impersonate police/CBI, keep an elderly victim on the line for hours, cut off their family, and drain them "to clear their name."

WHY NOTHING HELPS THE VICTIM

Every tool protects the institution, not the person on the call

- **Carrier / spam filters** block *numbers* — scammers rotate them in seconds.
- **Deepfake-voice detectors** ask "is the voice fake?" — useless when a *real human* reads the script, and they break on phone audio.
- **All of them are cloud services** — they'd have to stream a vulnerable person's private call to a server. Shady, and often illegal.
- And they show a warning **on a screen** — to someone who often **can't read it**. Half of India's illiterate population is over 50; ~11% of rural elders are digitally literate.

THE INSIGHT

Fight the part of the scam that never changes — the manipulation script

The voice keeps getting better. The playbook doesn't. Kavach detects the 8 **tactics** every scam uses:

Urgency

Secrecy

Untraceable payment

Authority impersonation

Distress hook

Isolation

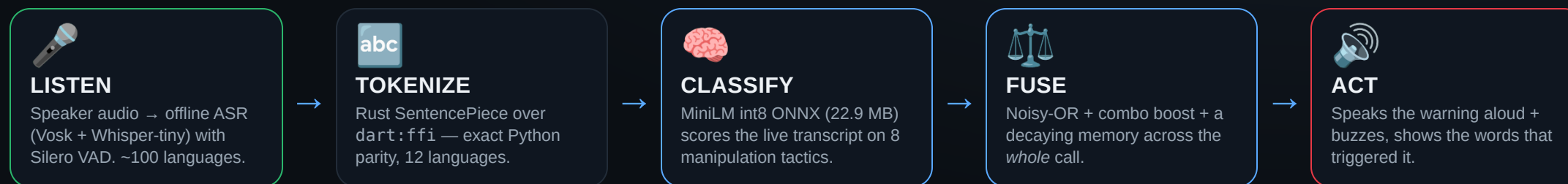
OTP / credential probe

Relationship spoof

A tiny int8 model scores the live transcript on-device, fuses the tactics into a risk level, and boosts dangerous combos (money + authority + isolation = the digital-arrest pattern).

HOW IT WORKS — END TO END

Five stages, all on one phone



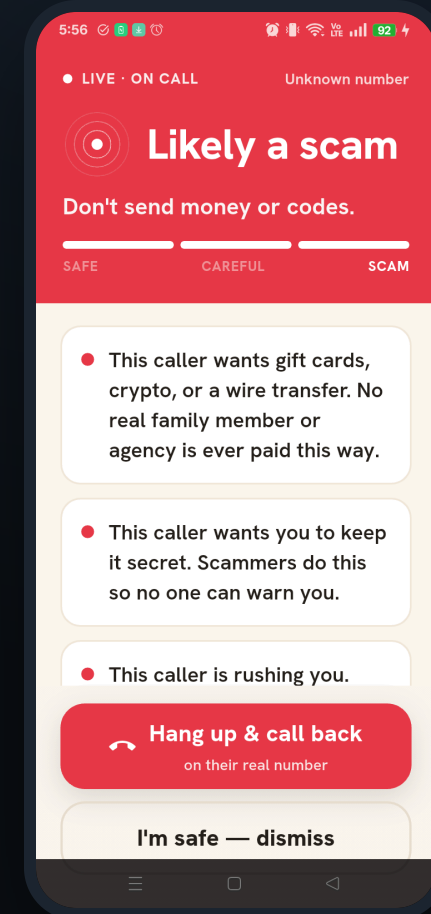
● Every step runs on-device · no INTERNET permission · the entire demo is in airplane mode

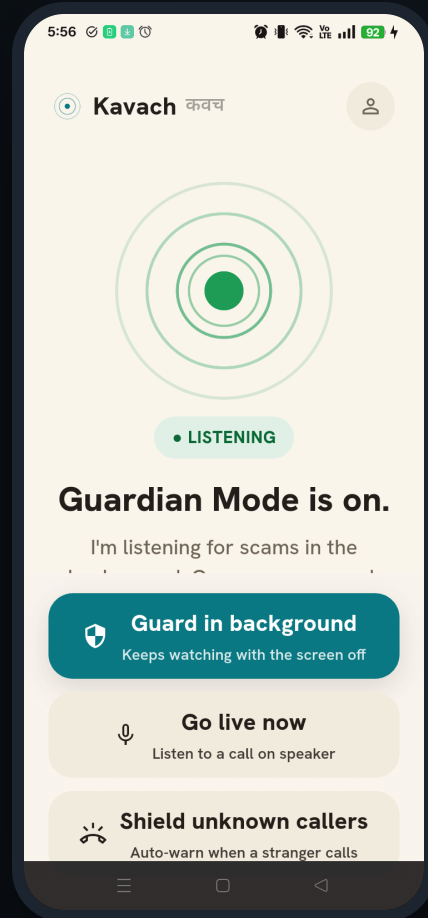
WHAT IT DOES

Follows the call live and calls it

- Listens on speaker, fully offline — SAFE...
- ...CAUTION as the pressure builds...
- ...**"Likely a scam"** — with the tactics in plain words and one action: **hang up & call back**.

Works two ways on the same engine: a live shield, and a background guardian with the screen off.





THE BREAKTHROUGH

It speaks up — for the person who can't read the screen

On HIGH, the phone says it **out loud**, in **the user's own language**, and **buzzes**:

"Stop. This may be a scam. Don't send money or share any code. Hang up and call your family."

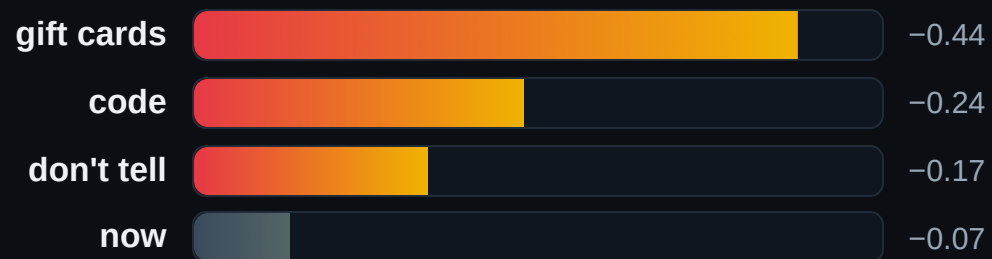
Colour + voice + vibration. Pre-recorded & offline — needs no TTS voice on the phone, can't hallucinate. **Validated on the OPPO A18.**

EXPLAINABLE — A REAL MODEL, NOT A SCRIPT

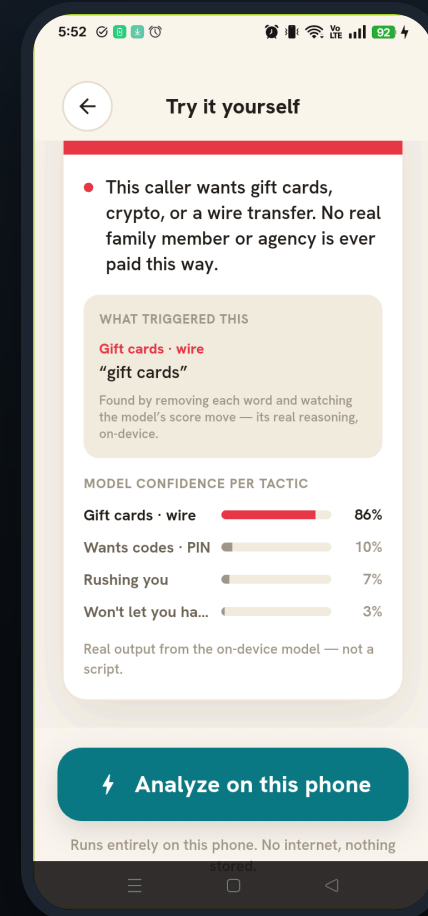
It shows the words it relied on

"buy **gift cards** and share the **code** now, don't tell anyone"

Occlusion attribution: remove each word → re-score on-device → rank by how far the risk drops.



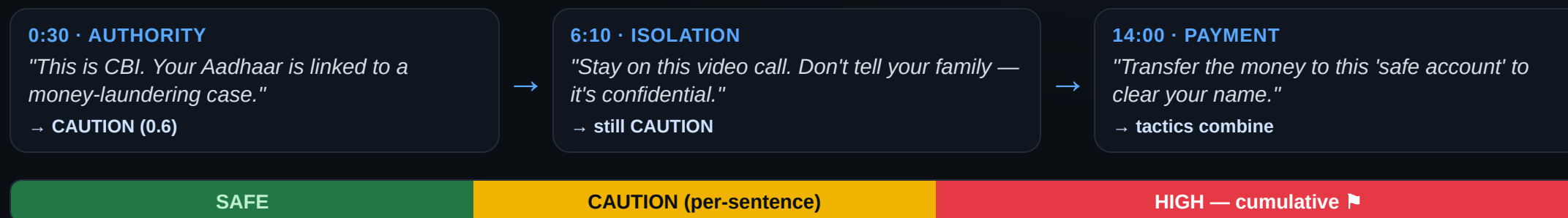
Per-tactic confidence is live model output — the user sees **why**, not a canned line.



THE HARD CASE — THE SLOW BURN

It remembers the whole call, not one sentence

A digital-arrest scam spreads the trap over 15 minutes. Each line alone looks only mildly off — so single-sentence detectors miss it. Kavach keeps a **decaying memory** of every tactic and re-fuses them, so the pattern still adds up.



On 1,378 real FTC robocalls this lifts recall from **28%** → **40%** (+43%) versus scoring one window at a time. Unit-tested on the slow-burn flip.

TECHNICAL IMPLEMENTATION

Honest engineering, measured on real fraud data

ARCHITECTURE

- Fine-tuned MiniLM, int8 ONNX 22.9 MB, on-device
- Rust **SentencePiece** tokenizer over `dart:ffi` → 12 languages
- Whisper-tiny + Silero VAD → **Tamil** + ~99 languages live
- **Conversation-level risk**: decaying memory across the whole call

MEASURED (FTC ROBOCALLS + BOTHBOSU)

40%

real-robocall recall vs 28% per-window (+43%)

99% / 65%

recall / precision (caller-side)

Advisory, never a blocker. We name the weak spots out loud (see EVALUATION.md).

PRIVACY BY ARCHITECTURE

Not a promise — a fact you can audit in one line

```
aapt dump permissions app-release.apk
```

→

```
android.permission.INTERNET count: 0
```

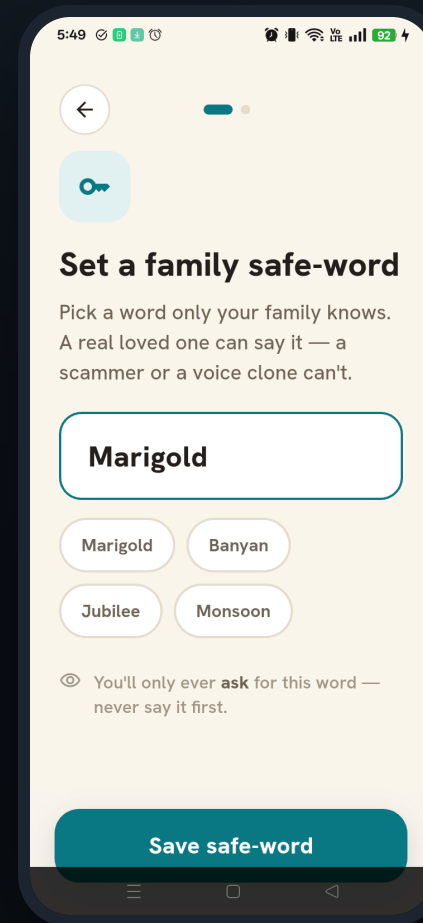
The release app declares **no INTERNET permission**. It **physically cannot** transmit your call — no cloud, no account, nothing stored. The entire demo runs in **airplane mode**.

Proof committed in the repo: `docs/RELEASE_PERMISSIONS.txt`

REAL-WORLD IMPACT & SCALABILITY

Built for the people who actually lose the money

- Runs offline on the **\$80 phones** the most-vulnerable users own
- **\$0 marginal cost** — no servers, no per-call inference. Scales to a billion phones
- Works in low-connectivity regions, and where streaming call audio is illegal
- Updates are model-file swaps; distribution is a normal app install



कवच

Catch the script.
Speak up. Before the
money's gone.

On-device · offline · multilingual · accessible ·
private by architecture. Validated on real
hardware.

GitHub: github.com/Anbu-00001/Kavach

Anbuchelvan Ganesan